

Chapter

Novel Cryptocurrency Investment Approaches: Risk Reduction and Diversification through Index Based Strategies

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Abstract

Cryptocurrency investment approaches continue to evolve rapidly. Traditionally, cryptocurrency holders tend to actively support up to several distinct projects, focusing their selection criteria on specific project characteristics, project team and community, existing markets and liquidity levels, as well as the perception of each unique project's broadly understood "mission and vision" and "future potential." In this chapter, we will explore an index-based investment strategy as an alternative to the more traditional single- or oligo-asset approaches. In the index-based paradigm, multi-asset strategy involves equalization and redistribution of risk exposure across multiple, pre-vetted portfolio positions. This strategy, novel to the cryptocurrency space, also involves risk reduction through cost averaging, dilution of cyber security-related risk(s), as well as mitigation of liquidity restrictions related to individual-position market liquidity characteristics. Additional discussion of software platforms, including both custodial and non-custodial wallets, and the associated risk-benefit considerations, will also be included in this manuscript.

Keywords: cryptocurrency, diversification, index, investing, risk modulation

1. Introduction

Cryptocurrencies are among the most recent entrants into the "investment vehicle" arena, dating back to the late 2000's and the development and introduction of Bitcoin (BTC) as the first and prototypical cryptographically secured asset [1, 2]. Ever since the introduction of early cryptocurrencies, it became evident that these largely experimental assets are prone to significant manipulation and volatility, not to mention scamming and other forms of fraud [3, 4]. Consequently, research into the various safety-focused and financially optimized approaches to "cryptographic asset investing" (CAI) came to the fore of the contemporary investment community, driven largely by early adopters [5–7]. As interest in CAI grows among mainstream investors around the globe, many questions naturally arise among potential new market participants, with key concerns increasingly in-line with traditionally "mainstream"



Figure 1.
A word cloud pictogram showing the high complexity of the overall cryptographic asset investment sphere. This asset space continues to evolve actively, with significant progress made since the 2010s, but a lot more work remaining to achieve full mainstream adoption.

issues (e.g., safekeeping, asset insurance, market liquidity and volatility, and other forms of guarantees) with each successive adoption cycle [8–10].

Among proposed strategies and approaches, index-based CAI emerged as one of the safest and less volatile options for long-term investors who are attracted by the potentially high returns associated with cryptographic assets, without the excessive exposure to risks associated with the traditionally high volatility of cryptocurrency markets [11–13]. In addition, index-based CAI (ibCAI) provides a degree of protection against any one component of the index itself becoming non-viable as an “investment vehicle,” either due to hacking or other factors, such as a numerically dwindling cryptocurrency community or lack of project-specific infrastructure/support [14–17]. There are multiple intersecting concepts involved within the overall cryptographic investment space, reflected in **Figure 1** pictogram. This chapter will discuss the importance and potential advantages of ibCAI, with an in-depth discussion of risks, benefits, and alternatives associated with this approach. Other key concepts discussed herein will include ibCAI storage strategies, inclusive of various types of software platforms (e.g., wallets and multi-wallets) required for the successful implementation of individualized ibCAI portfolios.

2. Cryptocurrency basics: the “currency versus investment” argument

Initially intended to serve as alternative (and often idealized) media of exchange [18–21], cryptocurrencies fairly quickly evolved and transformed to encompass various philosophies and different methodologies of decentralized ledger propagation and verification. Purposes behind individual cryptocurrencies vary widely, from socially-focused platforms to specialized data archival instruments, to purely financial-minded applications [22–24]. Consequently, a broad range of phenomena and behaviors have

been described in association with cryptocurrencies as “investment vehicles,” from various forms of data storage to conduits for mass-scale speculation and hoarding [25, 26].

Diversification within the cryptographic asset “investment vehicle” microcosm can be defined across numerous stratification categories. Perhaps the most important characteristic of each unique cryptographic asset is the way the actual decentralized ledger is propagated, from block to block [27, 28]. While the early pioneers, such as Bitcoin and Litecoin belonged to the so-called Proof-of-Work (PoW) family of cryptocurrencies [29, 30], the arrival of Peercoin introduced Proof-of-Stake (PoS) as an alternative [31, 32]. Further code development resulted in “hybrid” PoW and PoS block verification [33]. Further, diverse cryptography approaches have also been employed in parallel, with the emergence of multi-algorithm coins that provided greater overall blockchain security [34, 35].

3. The argument “in favor of” index-based cryptographic asset investment approach

Similar to traditional financial markets and other mainstream “investment vehicles,” cryptography-based assets exhibit certain characteristic (and at times predictable) features. In fact, analogous general approaches can apply to both individual cryptocurrencies and index-based vehicles alike. To begin with, ibCAI provides reduced minute-to-minute and day-to-day volatility [36]. Moreover, potential risk reduction benefits may also be present when cryptographic-based assets are included within a portfolio consisting of other asset classes (e.g., precious metals, commodities, real estate, fiat currencies, stocks, and bonds) [37, 38]. In the author’s personal observations, the smoothing effect of multi-class asset portfolios can substantially reduce overall volatility, but at the same time is less likely to result in above-average gains (and losses) characteristic of higher-risk, less diversified strategies.

It has long been known that long-term investment portfolio gains correlate with minimizing downside losses during market drops [39, 40]. It is also well established that market drops occur more abruptly (both in terms of time scales and price movements) than most market uptrends [41, 42]. Perhaps less known fact about market drops is that the overall “universe of assets” must account for both gains and losses, and that efficient market operations result in risk-based concentration of wealth (e.g., those who make risky bets in volatile markets, stand to gain more than those who are risk averse) [43, 44]. Consequently, a highly diverse portfolio, complete with defensive assets and non-traditional assets, such as cryptocurrencies and commodities, may help reduce the overall “downside risk” by more effectively capturing the flight of capital during significant market upheavals [45–48]. One can also reason (at least for most investors) that in the long run, “fewer overall losses” strategy will tend to outperform various “high gain / high loss” strategies, provided that the occasional “large losses” incurred in the latter are more likely to eclipse any “short-term gains” associated with high-risk bets.

Similar observations have been noted by our ibCAI expert group. More specifically, there are fairly well defined asset migration cycles between our large-cap, mid-cap, and small-cap portfolios. Such asset migration is highly reminiscent of stock market cycles based on indexed asset capitalization (small-, mid-, and large-caps) [49, 50]. For investors with shorter and medium-term horizons, this represents an excellent opportunity to “play the market” while reducing their overall

downside risk. At the same time, those who own a balanced portfolio of assets across all market capitalization categories, are able to enjoy the “long and steady” rise of their entire portfolio, without the need to worry about day-to-day asset class fluctuations [51–53].

Finally, additional important arguments “in favor of” a more diversified ibCAI portfolio have to do with liquidity and cybersecurity considerations. In terms of market liquidity, it is well established that low-liquidity markets can be easily overwhelmed by large-scale buy or sell orders, resulting in heightened volatility and limited options for market entry / exit tactics [54, 55]. Given the often brutal realities of low-liquidity cryptocurrency markets, investors have coined the term “pump and dump” where timing of market entry/exit must be very precise, or one will end up “holding the bag” once the post-rally market liquidity dries up [56, 57]. Asset holders within a well-balanced ibCAI portfolio are therefore more likely to be able to exit markets (at any given time) without triggering significant market declines due to inherently low market liquidity.

In terms of cybersecurity-based considerations, a more diversified cryptocurrency portfolio provides significant protection against single-position or even few-positions loss due to hacking. Under such circumstances, the diversity of blockchains within an individual portfolio, along with the equal distribution of funds across all of the participating assets, reduces the overall single-position risk to $(1/n)$ where “n” represents the total number of assets within the portfolio. In other words, a successful hack and a resultant loss of one of the positions in a portfolio of 30 positions will result in an overall loss of only 3.3%. This aspect of ibCAI portfolio management is often overlooked in favor of holding either a single or only a handful of assets.

4. Asset custody and reliability issues: know-your-customer (KYC) and other privacy related considerations

In the realm where privacy and confidentiality reign supreme, any approach to CAI is bound to meet the ever-present eyes of various regulatory agencies and entities responsible for taxation, legal and custody issues, and other aspects of financial market oversight [58–60]. For the purposes of this chapter, we will take a long-term view of traditional (non-speculative) investment approaches. In fact, the very concept of ibCAI represents the opposite of speculative, high-risk approaches of day-trading (e.g., time horizons of minutes to hours) and short-term (e.g., time horizons of <12 months) trading [61–63].

Inherent to the long-term stability and reliability of any “investment vehicle,” whether traditional or non-traditional, is one’s ability to reasonably closely follow the value and status of the portfolio/ holdings, as well as the general perception that the assets “will still be there” at the time of expected future redemption [64–66]. Thus, the issue of (legal) custody and associated systemic guarantees are central to such long-term stability. Moving further along this line of reasoning, the concept of “know-your-customer” (a.k.a., KYC) is also very important. After all, KYC-specific considerations (assuming that funds were initially procured in an honest/legitimate way) are among key protections the customer/holder has at his or her disposal to prove their ownership of a specific lot of cryptographic asset(s). In addition, KYC-based protections can also help prove that specific assets were lost to theft, even if investigations of said claim(s) are long and/or delayed [67–69].

5. Multi-wallets: the optimal individual investor approach?

Multi-wallet is a term that describes specialized software that is designed to securely hold multiple cryptocurrencies within a single shell or visual interface/container [70, 71]. Two broad categories of multi-wallets exist – Custodial and non-custodial. The former entails third-party control over end-user cryptocurrency private keys/account(s), with a well-defined set of vulnerabilities and potential downsides inherent to lack of direct end-user control [72, 73].

Non-custodial wallets, on the other hand, enable the end-user to fully control all aspects of the cryptographic asset custody and access considerations [74]. That said, there are important data security considerations, with significant vulnerabilities and opportunities for criminal exploitation of asset via both social engineering techniques and software hacking [70, 75]. Consequently, it is recommended at this time that individual users not only diversify their cryptocurrency portfolios, but also diversify the multi-wallets within which these portfolios are held.

With the introduction of simplified payment verification (SPV) methodology, transition from the more rigid architecture of “desktop-based wallets” (DBW) to “mobile-based multi-wallets” (MBMW) became possible [76, 77]. With the emergence of MBMWs, the end-user gained the power to essentially design his or her own, highly customized and personalized “do-it-yourself” (DIY) cryptocurrency portfolios. Such portfolios are complete with real-time price data inputs, user ability to determine and control asset ratios, as well as aggregate reporting of individual asset and portfolio values. Some software platforms, such as the Komodo Wallet, offer a multi-folio option, where several parallel portfolios can be created, each with its own set of private keys and secure password (Figure 2) [78, 79].



Figure 2.
An example of a software multi-wallet (Komodo Wallet) that constitutes a self-contained portfolio of cryptographic assets. Each asset is represented by its own line entry, complete with asset name, ticker symbol, current price, price change over a period of time, and other customized interface features.

Vaults and other multi-stakeholder custodial solutions have also been introduced, where assets are essentially “irretrievably locked” until a certain pre-defined number of authorized users “agree in principle” to release such assets to a third-party address [80]. This approach represents an attractive option for multi-person entities, such as families or small investment clubs, where assets can only be moved when a pre-determined number of participants all agree on a specific transfer operation. The multi-signature “vault storage” approach has also been popular among owners of various high-value cryptographic assets (such as non-fungible tokens or NFTs) [81, 82].

6. Technical aspects of index-based approaches

The overall diversification of the blockchain/cryptocurrency space resulted in significant diversification of customized code that allowed alternative cryptocurrencies (e.g., “AltCoins”) to proliferate within the early (2009–2017) phases of the cryptographic asset era [83, 84]. In the context of asset holding, the logistics of diversification became formidable because of the requirement for cryptocurrency-specific “wallets” to be deployed for each unique asset. Two durable alternatives emerged to help solve this conundrum – the previously outlined implementations of SPV [77] and asset tokenization (AT) [85]. The latter, in general, relies on a primary blockchain (such as Ethereum or Quantum) to provide an “entry point” for different token-assigned value units (e.g., a single token represents a pre-defined item, a primary asset, or some other quantity, etc.) that is then uniquely identified under its own “contract address” (or essentially a “subaccount” that then holds secondary assets/tokens on the primary blockchain).

At this time, both custodial and non-custodial approaches to ibCAI management exist, both for different blockchains and for various unique tokens/tokenized assets. As outlined in previous sections, each of the approaches, along with some other associated nuances, may have a unique risk-benefit profile. Each end-user is encouraged to carefully consider all of the available options before proceeding. Factors that need to be taken into consideration include one’s technical/computer/blockchain expertise, financial risk tolerance, the choice between active or passive management styles, the balance between privacy versus convenience, and the availability of specialized adjunctive financial products (such as “asset loss” insurance policies) [86, 87]. Finally, one must consider portfolio entry/exit points, such as either centralized or decentralized cryptocurrency exchanges. After all, most wallets only “hold assets” but are not able to facilitate active on/off-boarding of said assets. This aspect of ibCAI management also requires careful and comprehensive evaluation of risks, benefits, and alternatives by the end-user.

7. Conclusion

Cryptographic assets are here to stay, and so is cryptographic asset investing (CAI). Within this greater universe of asset management options, index-based CAI represents an emerging and increasingly popular option for volatility and risk reduction, with multiple potential advantages when compared to single- or oligo-asset CAI strategy. More specifically, the ibCAI approach minimizes the magnitude of any single-asset loss due to hacking or other exploits. It also helps facilitate portfolio off-loading in the setting of low-liquidity markets. Asset custody and software-based

conduits for secure storage of cryptocurrencies and cryptocurrency portfolios continue to evolve, and represent a significant potential risk/attack surface. The area of ibCAI is likely to evolve and mature over the coming years, with emerging legal, KYC, and tax frameworks that will help introduce this approach to cryptocurrency investment into the mainstream.

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References

- [1] Nakamoto S. Bitcoin: A peer-to-peer electronic cash system. *Decentralized Business Review*. 2008;1(1):1-9
- [2] Stawicki SP. Introductory chapter: Early applications and evolution of Blockchain implementations in medicine. In: *Blockchain in Healthcare: From Disruption to Integration*. Cham, Switzerland: Springer; 2023. pp. 1-5
- [3] Barnes P. Crypto currency and its susceptibility to speculative bubbles, manipulation, scams and fraud. *Journal of Advanced Studies in Finance (JASF)*. 2018;9((18)):60-77
- [4] Twomey D, Mann A. Fraud and manipulation within cryptocurrency markets. *Corruption and Fraud in Financial Markets: Malpractice, Misconduct and Manipulation*. 2020:205-249
- [5] Petukhina A, Trimborn S, Härdle WK, Elendner H. Investing with Cryptocurrencies-evaluating their potential for portfolio allocation strategies. *arXiv preprint arXiv:2009.04461*. 9 Sep 2020
- [6] Elendner H. F5: Optimised cryptocurrency investment strategies. White paper, 2018. Available from: https://f5crypto.com/wp-content/uploads/2023/06/F5_Crypto_Index_Whitepaper-1.pdf
- [7] Andrianto Y, Diputra Y. The effect of cryptocurrency on investment portfolio effectiveness. *Journal of Finance and Accounting*. 2017;5(6):229-238
- [8] Spenkelink H. The Adoption Process of Cryptocurrencies-Identifying Factors That Influence the Adoption of Cryptocurrencies from a Multiple Stakeholder Perspective. Enschede, Netherlands: University of Twente; 2014;1:1-103
- [9] Saiedi E, Broström A, Ruiz F. Global drivers of cryptocurrency infrastructure adoption. *Small Business Economics*. 2021;57:353-406
- [10] Alzahrani S, Daim TU. Analysis of the cryptocurrency adoption decision: Literature review. In: *2019 Portland International Conference on Management of Engineering and Technology (PICMET)*. Portland, Oregon: IEEE; 2019;1:1-11
- [11] Hu Y, Rachev ST, Fabozzi FJ. Modelling crypto asset price dynamics, optimal crypto portfolio, and crypto option valuation. 2019. *arXiv preprint arXiv:1908.05419*
- [12] Trimborn S, Härdle WK. CRIX an index for cryptocurrencies. *Journal of Empirical Finance*. 2018;49:107-122
- [13] EWMCI. East-West Magnate Cryptocurrency Investments. 2023 [cited 2023 November 30]. 2023 Available from: <https://ewmci.info/>
- [14] Grobys K, Sapkota N. Predicting cryptocurrency defaults. *Applied Economics*. 2020;52(46):5060-5076
- [15] Stylianou K et al. Cryptocurrency competition and market concentration in the presence of network effects. *Ledger*. 2021;6:81-101
- [16] Howson P, de Vries A. Preying on the poor? Opportunities and challenges for tackling the social and environmental threats of cryptocurrencies for vulnerable and low-income communities.

Energy Research & Social Science.
 2022;**84**:102394

[17] Trump BD et al. Cryptocurrency: Governance for what was meant to be ungovernable. *Environment Systems and Decisions*. 2018;**38**:426-430

[18] Brunton F. *Digital Cash: The Unknown History of the Anarchists, Utopians, and Technologists Who Created Cryptocurrency*. Princeton, New Jersey: Princeton University Press; 2020

[19] Shapiro M. Crypto-egalitarian life: Ideational and materialist approaches to Bitcoin. *Social Analysis*. 2022;**66**(3):62-82

[20] Petz M, Eskelinen T. The politics of money utopias: Methodological utopian explorations of the Banjar and the Neocracy. *International Journal of Community Currency Research*. 2022;**26**(1):1-21

[21] Jóhannsson J. *Sharing Economy Platforms: A Path towards Progress or Inequality?*; 2017

[22] Ba CT, Zignani M, Gaito S. The role of cryptocurrency in the dynamics of blockchain-based social networks: The case of Steemit. *PLoS One*. 2022;**17**(6):e0267612

[23] Shrivastava G, Le D-N, Sharma K. *Cryptocurrencies and Blockchain Technology Applications*. Hoboken, New Jersey: John Wiley & Sons; 2020

[24] Stawicki S. *Blockchain in Healthcare: From Disruption to Integration*. Vol. 10. Cham, Switzerland: Springer Nature; 2023

[25] Vejačka M. Basic aspects of cryptocurrencies. *Journal of Economy, Business and Financing*. 2014;**2**(2):75-83

[26] Pernice IGA, Almosova A, Elendner H. Cryptocurrency: Speculative asset and medium of exchange. 2021. SSRN 3964372. Available from: https://papers.ssrn.com/sol3/Delivery.cfm/SSRN_ID4069505_code3039903.pdf?abstractid=3964372&mirid=1

[27] Stawicki SP, Firstenberg MS, Papadimos TJ. What's New in Academic Medicine? Blockchain Technology in Health-Care: Bigger, Better, Fairer, Faster, and Leaner. *Medknow*; 2018. pp. 1-11

[28] Aggarwal S, Kumar N. Cryptographic consensus mechanisms. In: *Advances in Computers*. Philadelphia, Pennsylvania: Elsevier; 2021. pp. 211-226

[29] Auer R. *Beyond the Doomsday Economics of 'Proof-of-Work' in Cryptocurrencies*. Basel, Switzerland: Bank of International Settlements; 2019

[30] Capponi A, Olafsson S, Alsabah H. Proof-of-work cryptocurrencies: Does mining technology undermine decentralization? *Management Science*. 2023;**6**:1-53

[31] Bentov I, Gabizon A, Mizrahi A. Cryptocurrencies without proof of work. In: *International Conference on Financial Cryptography and Data Security*. Cham, Switzerland: Springer; 2016

[32] Zhao W et al. On peercoin proof of stake for blockchain consensus. In: *2021 The 3rd International Conference on Blockchain Technology*. Kuala Lumpur, Malaysia. 2021;**3**:129-134

[33] Chepurnoy A. et al., Twinscoin: A cryptocurrency via proof-of-work and proof-of-stake. *Cryptology ePrint Archive*. 2017. Available from: <https://eprint.iacr.org/2017/232.pdf>

- [34] Lovejoy JPT. An Empirical Analysis of Chain Reorganizations and Double-Spend Attacks on Proof-of-Work Cryptocurrencies. Boston, Massachusetts: Massachusetts Institute of Technology; 2020
- [35] Arab Monetary Fund. Abu Dhabi, United Arab Emirates: Digital Currency Technologies; 2021;23:1-37
- [36] Chabakauri G, Rytchkov O. Asset pricing with index investing. *Journal of Financial Economics*. 2021;141(1):195-216
- [37] Čiumakovas I. Diversifying Your Portfolio with Cryptocurrency-Does it Improve Sharpe Ratios for Institutional Investors?. Enschede, Netherlands: University of Twente; 2022
- [38] Rudolf KO, El Zein SA, Lansdowne NJ. Bitcoin as an investment and hedge alternative. A DCC MGARCH model analysis. *Risks*. 2021;9(9):154
- [39] Jorion P. Risk management lessons from long-term capital management. *European Financial Management*. 2000;6(3):277-300
- [40] Johansen A, Sornette D. Shocks, crashes and bubbles in financial markets. *Brussels Economic Review*. 2010;53(2):201-253
- [41] Fama EF. The behavior of stock-market prices. *The Journal of Business*. 1965;38(1):34-105
- [42] Bond FD. *Stock Prices: Factors in their Rise and Fall, Illustrated with Charts*. Vol. 6. New York, New York: Moody's Magazine Book Department; 1911
- [43] Sauer RD. The economics of wagering markets. *Journal of Economic Literature*. 1998;36(4):2021-2064
- [44] Fong W. *The Lottery Mindset: Investors, Gambling and the Stock Market*. Cham, Switzerland: Springer; 2014
- [45] Hyung N, De Vries CG. Portfolio diversification effects of downside risk. *Journal of Financial Econometrics*. 2005;3(1):107-125
- [46] Bhansali V, Holdom J. Diversifying diversification: Downside risk management with portfolios of insurance securities. *The Journal of Portfolio Management*. 2021;47(6):101-113
- [47] Adamiec L, Cernauskas D. Optimal number of assets for reduction of market risk through diversification. *International Journal of Economics, Business and Management Research*. 2019;3:45-54
- [48] Bouri E, Shahzad SJH, Roubaud D. Cryptocurrencies as hedges and safe-havens for US equity sectors. *The Quarterly Review of Economics and Finance*. 2020;75:294-307
- [49] Copeland MM, Copeland TE. Market timing: Style and size rotation using the VIX. *Financial Analysts Journal*. 1999;55(2):73-81
- [50] Lucas A, Van Dijk R, Kloek T. Stock selection, style rotation, and risk. *Journal of Empirical Finance*. 2002;9(1):1-34
- [51] Senthilkumar A, Namboothiri A, Rajeev S. Does portfolio optimization favor sector or broad market investments? *Journal of Public Affairs*. 2022;22:e2752
- [52] Gupta A, Subramanian RA. *Harvesting Sector Beta through Broad Sector Indices*. Paris, France: MSCI Index Research; 2013
- [53] Bender J et al. Portfolio of risk premia: A new approach to

diversification. *The Journal of Portfolio Management*. 2010;**36**(2):17-25

Journal of Sustainable Finance & Investment. 2019;**9**(4):356-377

[54] Baxter G, Cartlidge J. Flying by the seat of their pants: What can high frequency trading learn from aviation? In: *Proceedings of the 3rd International Conference on Application and Theory of Automation in Command and Control Systems*. Naples, Italy; 2013

[63] van Loon RJ. Long-term investing and the frequency of investment decisions. *The Journal of Portfolio Management*. 2021;**47**(8):86-104

[55] Clunie J. *Predatory Trading and Crowded Exits: New Thinking on Market Volatility*. Petersfield, Hampshire, UK: Harriman House Limited; 2010

[64] Siegel JJ. *Stocks for the Long Run: The Definitive Guide to Financial Market Returns & Long-Term Investment Strategies*. New York, New York: McGraw-Hill Education; 2021

[56] Xu J, Livshits B. The anatomy of a cryptocurrency {Pump-and-Dump} scheme. In: *28th USENIX Security Symposium (USENIX Security 19)*. Santa Clara, California, USA; 2019

[65] Gorton G, Winton A. Financial intermediation. In: *Handbook of the Economics of Finance*. Philadelphia, Pennsylvania: Elsevier; 2003. pp. 431-552

[57] Dhawan A, Putniņš TJ. A new wolf in town? Pump-and-dump manipulation in cryptocurrency markets. *Review of Finance*. 2023;**27**(3):935-975

[66] Gelpern A, Gerding EF. Inside safe assets. *Yale Journal of Register*. 2016;**33**:363

[58] Peláez-Repiso A, Sánchez-Núñez P, Calvente YG. Tax regulation on blockchain and cryptocurrency: The implications for open innovation. *Journal of Open Innovation: Technology, Market, and Complexity*. 2021;**7**(1):98

[67] Kuperberg M. Blockchain-based identity management: A survey from the enterprise and ecosystem perspective. *IEEE Transactions on Engineering Management*. 2019;**67**(4):1008-1027

[59] Aziz A. Cryptocurrency: Evolution & legal dimension. *International Journal of Business, Economics and Law*. 2019;**18**(4):31-33

[68] Tironsakkul T. et al. Probing the mystery of cryptocurrency theft: An investigation into methods for taint analysis. 2019. arXiv preprint arXiv:1906.05754

[60] Martinčević I et al. Accounting and tax regulation of cryptocurrencies. *Interdisciplinary Description of Complex Systems: INDECS*. 2022;**20**(5):640-661

[69] Bischooping G. Prosecuting cryptocurrency theft with the defend trade secrets act of 2016. *University of Law Review*. 2018;**167**:239

[61] Jacobs BI, Levy KN, Starer D. Long/short equity investing. *Journal of Portfolio Management*. 1993;**20**(1):52

[70] Shi C, Guan Y. Forensic analysis of android cryptocurrency wallet applications. In: *IFIP International Conference on Digital Forensics*. Cham, Switzerland: Springer; 2023

[62] Schoenmaker D, Schramade W. Investing for long-term value creation.

[71] Lan R. et al. Horizon: A gas-efficient, trustless bridge for cross-chain transactions. 2021. arXiv preprint arXiv:2101.06000

- [72] Au A, et al. Care, Custody, & Control (CCC): Identification, Quantification, and Mitigation of Cryptocurrency Custodial Risk: Cryptocurrency and Cyber Risk: Market Analysis and Perspectives. New York; 2020
- [73] Barbereau T, Bodó B. Beyond financial regulation of crypto-asset wallet software: In search of secondary liability. *Computer Law & Security Review*. 2023;**49**:105829
- [74] Sans T, Liu Z, Oh K. A decentralized mnemonic backup system for non-custodial cryptocurrency wallets. In: *International Symposium on Foundations and Practice of Security*. Cham, Switzerland: Springer; 2022
- [75] Kushnirenko SP, Kharatishvili AG. Cryptocurrencies turnover and forensic analysis of the mechanism of committing crimes. *Kutafin Law Review*. 2023;**9**(4):774-792
- [76] Faridi A, Siddiqui F. Improving SPV-based cryptocurrency wallet. In: *Cybernetics, Cognition and Machine Learning Applications: Proceedings of ICCMLA 2019*. Cham, Switzerland: Springer; 2020
- [77] Biryukov A, Tikhomirov S. Security and privacy of mobile wallet users in Bitcoin, dash, Monero, and Zcash. *Pervasive and Mobile Computing*. 2019;**59**:101030
- [78] Beutin N, Boran D. *The Great Web 3.0 Glossary*. Frankfurt am Main, Germany: Deutscher Fachverlag GmbH; 2023
- [79] Kiechl P. *Extension and Standardization of a Blockchain Interoperability API*. Switzerland: Zürich; 2021
- [80] Swambo J. et al. Custody protocols using bitcoin vaults. 2020. arXiv preprint arXiv:2005.11776
- [81] Das D et al. Understanding security issues in the NFT ecosystem. In: *Proceedings of the 2022 ACM SIGSAC Conference on Computer and Communications Security*. Copenhagen, Denmark, EU; 2022
- [82] Ruan X. *Exploring Vulnerabilities and Anomalies in NFT Marketplaces*. Guelph, Ontario, Canada: University of Guelph; 2022
- [83] Ciaian P, Rajcaniova M. Virtual relationships: Short-and long-run evidence from BitCoin and altcoin markets. *Journal of International Financial Markets, Institutions and Money*. 2018;**52**:173-195
- [84] Bousfield D. Crypto-coin hierarchies: Social contestation in blockchain networks. *Global Networks*. 2019;**19**(3):291-307
- [85] Caldarelli G. Wrapping trust for interoperability: A preliminary study of wrapped tokens. *Information*. 2021;**13**(1):6
- [86] Sun M. A breakthrough of digital assets security in crypto insurance: Cyber risk prediction models. *Research and applications towards mathematics and computer*. Science. 2023:54
- [87] Zuckerman A. *Insuring crypto: The birth of digital asset insurance*. University of Illinois Journal of Law, Technology and Policy. 2021:75-119